

WEB DEVELOPMENT - TASK 3

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Overview:

In this task, I first developed a working authentication system to understand the basics of express, node js, and mongoose. After that, I used AI to generate a senior-level version of the same code. This report presents my personal findings and the differences I observed between my original implementation and the AI-generated code.

Code Maintainability and Readability:\

- **My Code:** I used concise variable names like `u` and `result`, which made the logic easy for me to follow as a solo developer during the learning phase.
- **AI Implementation:** The AI used more descriptive and self-documenting naming conventions, such as `userInstance` and `registrationResult`.

Security Evaluation:

The most significant difference I noticed was in password storage.

- **My Implementation:** I stored passwords in plain text, which I later realized is not a secure practice.
- **AI Implementation:** The AI used `bcryptjs` to hash and store passwords. I had to manually install this package in order to run the AI-generated code, and it helped me understand how sensitive data is protected in real-world applications.

Route Versioning:

- **My Implementation:** I used basic routes such as `/login` and `/register`. Additionally, my error handling was general.
- **AI Implementation:** The AI version used versioned API routes such as `/api/v1`. This was a new concept for me, but I understand that this practice helps with future-proofing the code. It also used specific HTTP status codes such as 409 and 201 instead of simply returning text responses.

Session Handling:

- **My Implementation:** I used a simple boolean check for sessions
- **AI Implementation:** The AI treated the session as a structured data object. It also added security flags like `httpOnly` and `maxAge` to the cookies which makes the session much more secure.

Conclusion:

Comparing my code to AI's code showed me that while my version is beginner friendly and functional , the AI version is closer to being industry ready. The AI code is more structured, handles error more specifically, and places greater emphasis on security. This task helped me realize that making code work is different than making code professional.